

Cross-Circuit Formula 1 Telemetry Performance Decomposition

When an analytic baseline outperforms the learned model

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Research artifact from the RaceCast telemetry analysis pipeline | August 2026 | Version 2.0

github.com/ProjectsofSadia/RaceCast-Research

Abstract

This study evaluates a public-telemetry workflow for decomposing Formula 1 qualifying-lap performance into local track windows, and tests whether machine learning adds anything beyond a direct analytic calculation. Data from five 2024 qualifying sessions were filtered under explicit quality rules and converted into 372 same-driver lap comparisons and 2,487 adaptive segment observations.

The original gradient-boosting model reached a mean leave-one-circuit-out MAE of 0.0563 s and mean R^2 of 0.7258. A non-learned inverse-speed reconstruction, derived from segment length, reference segment time and mean-speed delta, beat it: mean MAE 0.0408 s and mean R^2 0.8127. After subtracting the analytic reconstruction, a control and powertrain model failed to predict the residual on every held-out circuit, with mean R^2 of -0.275.

The initial machine-learning formulation was therefore approximating a relationship already available analytically from speed and distance, and the public control channels evaluated here carried no transferable residual signal. The contribution is methodological. The work documents an open telemetry-decomposition pipeline, demonstrates why strong analytic baselines matter, and identifies concrete failure modes around target coupling, scale-dependent evaluation, adaptive segmentation and public-data limits.

Keywords: Formula 1; telemetry; lap-time decomposition; motorsport analytics; analytic baseline; machine learning; cross-circuit validation; negative result

1. Engineering Motivation and Research Objectives

An overall lap-time delta tells you how much faster one lap was. It does not tell you where the difference accumulated, or whether a learned model is needed to explain it. I built RaceCast to turn public telemetry into localized comparisons of speed, braking, throttle, RPM, gear and the other available channels.

The initial question was whether segment-level time loss could be predicted on a circuit excluded from training. A methodological audit then showed that the target, traversal-time difference, is mathematically linked to the speed-derived predictors. That finding forced a stricter question: does machine learning add transferable information beyond the analytic speed-time relationship already sitting in the telemetry?

The revised research questions are:

- RQ1: How accurately can a direct analytic inverse-speed reconstruction reproduce segment-level time differences across held-out circuits?
- RQ2: Do the learned models outperform simple non-learned baselines under the same circuit-held-out evaluation?
- RQ3: After removing the analytic speed-based component of the target, do public control and powertrain features predict the remaining residual across circuits?

- RQ4: How do target scale, adaptive segmentation, selection rules and public-telemetry limits affect interpretation?

2. Data and Selection

The dataset was extracted through FastF1 3.8.3 from 2024 Formula 1 qualifying sessions at Bahrain, Monaco, Silverstone, Monza and Suzuka. Candidate laps had to carry a valid lap time, pass FastF1's IsAccurate flag where available, show no pit-in or pit-out marker, and satisfy FastF1 pick_quicklaps at a 1.07 threshold. I applied no additional outlier trimming.

That rule produced 469 selected laps. Telemetry extraction succeeded for 467. Two failed because telemetry processing returned an empty sequence: ALB lap 28 at Monaco and MAG lap 2 at Suzuka. The final comparison set holds 372 same-driver lap pairs.

Circuit	Selected laps	Drivers	Pairs
Bahrain	85	20	65
Monaco	143	20	122
Monza	87	20	67
Silverstone	80	15	65
Suzuka	74	20	53

The 1.07 quick-lap rule left only 15 represented drivers at Silverstone against 20 everywhere else. I report that as a consequence of the filter and do not infer a cause. Across all pairs, 98.9% used the same tyre compound.

Official lap-time deltas had a median of 0.661 s, a 95th percentile of 3.228 s and a maximum of 5.566 s, so the dataset contains substantially compromised laps. I kept them and treat their presence as a limitation rather than trimming them out to make the numbers look better.

3. Telemetry Alignment and Adaptive Segmentation

Telemetry was aligned to a 5 m distance grid. One naming quirk needs stating: the compact file delta_time_50m.csv stores every tenth aligned sample for visualization, so its exported spacing is 50 m while the analysis itself runs at 5 m. Local traversal time was approximated as $\Delta d/v$ after converting speed to m/s, and the cumulative comparison-to-reference delta came from summing local traversal-time differences.

For each driver and event, the fastest selected telemetry-available lap served as the reference, and local windows were detected from speed minima on that driver's reference lap. This is adaptive segmentation, not a canonical corner map, and it did not behave uniformly.

Circuit	Segments per driver	Max center SD (m)
Bahrain	8	8

Circuit	Segments per driver	Max center SD (m)
Monaco	7	8
Monza	6	11
Silverstone	5, 6	286
Suzuka	6, 7	519

Bahrain, Monaco and Monza produced stable segment counts with small center dispersion across drivers. Silverstone and Suzuka did not, reaching maximum segment-center standard deviations of roughly 286 m and 519 m. This revision therefore does not treat a shared segment_id as the same corner across drivers at those two circuits. Each row is a local performance window, and fixed circuit-level segmentation is left for future work.

4. Models, Baselines and Validation

The original primary analysis compared linear regression, random forest regression and histogram gradient boosting. The configured gradient-boosting model used `learning_rate=0.06`, `max_iter=350`, `max_leaf_nodes=31`, `l2_regularization=0.5` and `random_state=42`.

Evaluation used leave-one-circuit-out validation. All observations from one circuit were withheld, the model trained on the other four, and the process repeated for all five. The original feature set included speed, braking, throttle, exit-acceleration, RPM, gear, DRS, segment length, reference segment time, and tyre-life delta in the headline model. The earlier audit table omitted tyre-life delta. I disclose that discrepancy here rather than presenting the two as the same run.

This revision adds three mandatory non-learned baselines: predicting zero, predicting the training-target mean, and an analytic inverse-speed reconstruction. The analytic estimate is:

$$v_{\text{ref}} = L / t_{\text{ref}}; \quad v_{\text{cmp}} \approx v_{\text{ref}} + \Delta v_{\text{mean}}; \quad \Delta t_{\text{analytic}} = L / v_{\text{cmp}} - t_{\text{ref}}$$

where L is segment length, t_{ref} is reference segment time, and Δv_{mean} is the mean-speed difference converted to m/s. The formula fits no parameters.

5. Results

5.1 Learned models versus baselines

Method	Mean MAE (s)	Fold SD MAE	Mean R ²
Analytic reconstruction	0.0408	0.0096	0.8127
Gradient Boosting	0.0563	0.0310	0.7258
Random Forest	0.0575	0.0335	0.6965
Linear Regression	0.0701	0.0373	0.6261
Predict training mean	0.1330	0.0611	-0.2975
Predict zero	0.1570	0.0890	-0.7129

The analytic reconstruction is the best method in the evaluated set. The principal result is therefore negative for the machine-learning hypothesis: the learned model does not earn its complexity when the target can be reconstructed more accurately, and with no fitted parameters, straight from the speed-time relationship.

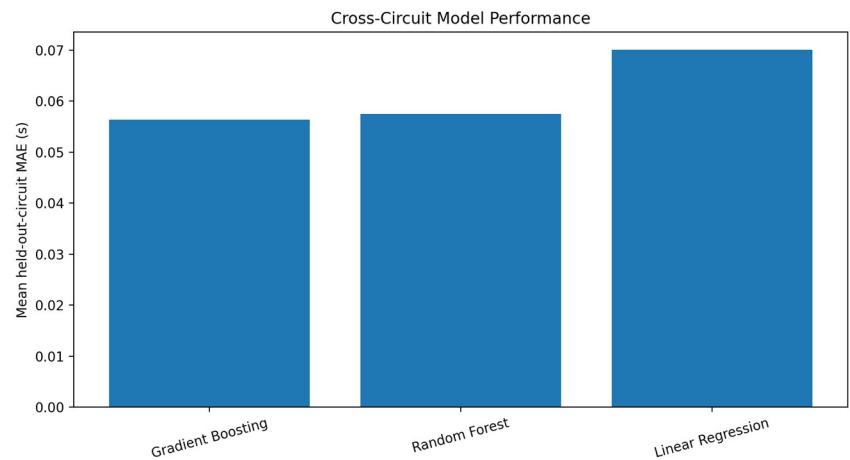


Figure 1. Cross-circuit performance of the analytic reconstruction, the machine-learning models, and the naive reference baselines under leave-one-circuit-out validation, reported as mean absolute error.

5.2 Circuit-level performance and scale normalization

Circuit	GB MAE	GB R ²	Analytic MAE	Analytic R ²	GB MAE/MAD
Bahrain	0.0344	0.7132	0.0328	0.7520	0.600
Monaco	0.0619	0.6765	0.0565	0.7395	0.765
Monza	0.0449	0.7607	0.0395	0.8891	0.839
Silverstone	0.1078	0.7085	0.0419	0.9528	0.735
Suzuka	0.0327	0.7700	0.0333	0.7303	0.540

Raw MAE is scale dependent, which changes the reading of this table. Silverstone shows the largest raw gradient-boosting MAE at 0.1078 s, but its segment-time deltas are also much larger. Normalizing by each circuit's target median absolute deviation puts Monza at the highest MAE/MAD ratio, not Silverstone. I withdraw the earlier interpretation that Silverstone demonstrated uniquely poor transfer.

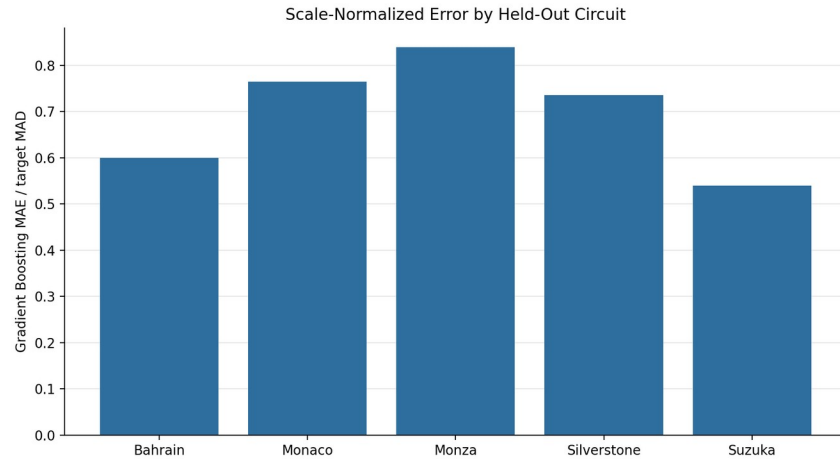


Figure 2. Circuit-specific gradient boosting error normalized by the median absolute deviation of the segment-time-delta target, giving a scale-adjusted comparison across held-out circuits.

5.3 Residualized experiment

Held-out circuit	n	Residual MAE	Residual R ²
Bahrain	520	0.0360	-0.2770
Monaco	854	0.0575	-0.1038
Monza	402	0.0383	-0.0921
Silverstone	385	0.0567	-0.5906
Suzuka	326	0.0350	-0.3137

After subtracting the analytic reconstruction, a gradient-boosting model using braking, throttle, RPM, gear, DRS, segment length and reference segment time produced negative R² on every held-out circuit, mean - 0.275. Negative R² means the model performed worse than predicting the residual mean. Within this formulation, the evaluated control and powertrain features supply no transferable residual predictive signal beyond the analytic speed-based component.

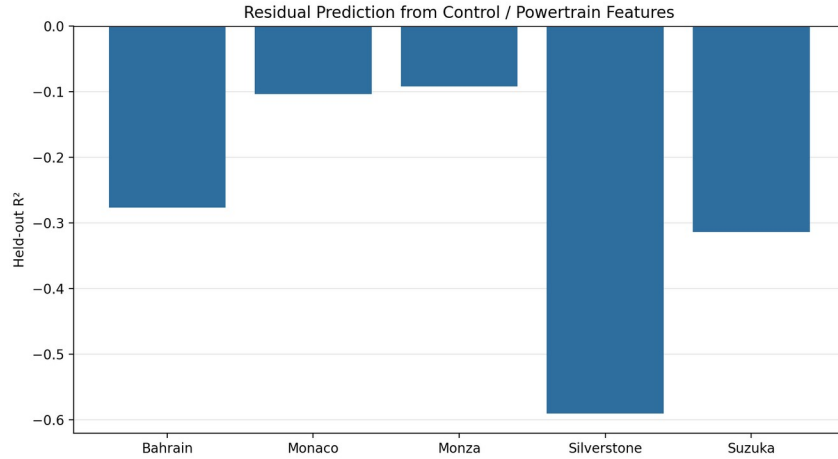


Figure 3. Leave-one-circuit-out R^2 for residual segment-time prediction using control and powertrain features, after removal of the analytic speed-time component. Every circuit is negative.

5.4 Target distribution, timing fidelity and outliers

The 2,487 segment deltas have mean 0.141 s, median 0.101 s, standard deviation 0.202 s, minimum -0.290 s, 95th percentile 0.495 s, 99th percentile 0.921 s and maximum 2.731 s. 14.23% are negative.

Telemetry-integrated lap delta correlates with official lap delta at $r=0.9937$ across the full range, but that falls to $r=0.9428$ once the comparison is restricted to lap deltas under 1 s. Integration carries a mean bias of +0.0119 s and a mean absolute discrepancy of 0.0846 s against official lap timing. The model predicts a constructed telemetry target, not official timing, so sub-0.1-s model errors should not be read as equivalent physical accuracy against official lap time.

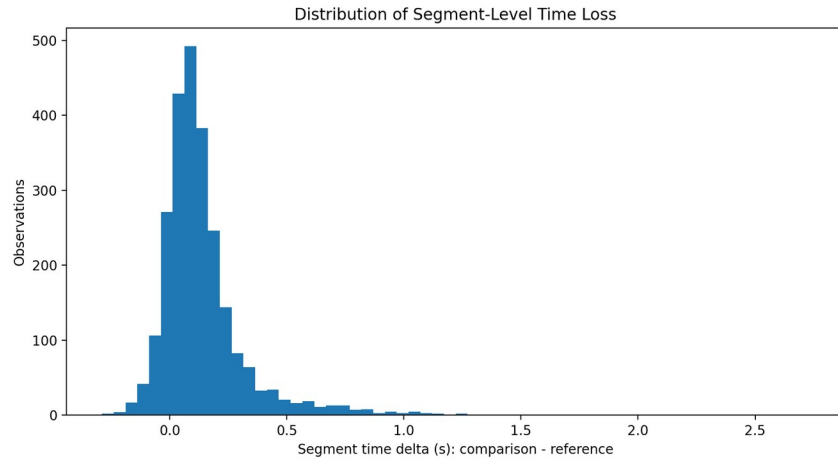


Figure 4. Empirical distribution of constructed segment-level traversal-time differences across the five qualifying-event datasets.

5.5 Feature ablation and explainability

The original ablation table survives only as descriptive evidence. Gradient boosting fold-to-fold MAE standard deviation is 0.0310 s. The changes from removing braking (-0.00035 s), throttle (-0.00054 s), exit-

acceleration (+0.00041 s) or powertrain (-0.00281 s) all sit an order of magnitude below that cross-circuit variation, so I do not interpret them as meaningful.

Removing the corner-speed group increased mean MAE by 0.0273 s, which is expected given that speed is algebraically linked to traversal time. SHAP and permutation-importance graphics from the original model are therefore model-behavior diagnostics. They are not evidence of physical causality or of independent feature value.

6. Discussion

The revised result is simpler and stronger than the original story. RaceCast localizes performance differences and organizes public telemetry into analyzable windows, but the specific machine-learning target used here does not justify a learned model. A direct inverse-speed calculation is both simpler and more accurate.

This is a common applied-ML failure mode. When a target is constructed from a physical quantity already present in the features, a high R^2 can reflect algebraic reconstruction instead of discovery. The residualized experiment sharpens the point: once the analytic speed component is removed, the available public control and powertrain channels fail to predict what is left across unseen circuits.

That does not mean braking, throttle, RPM or gear are physically unimportant. It means that in this representation, at public telemetry resolution, they contribute no stable cross-circuit signal that improves prediction of the residual target.

The value of the work is methodological and tooling-oriented: explicit distance alignment, transparent lap selection, within-driver comparisons, circuit-held-out evaluation, analytic baselines, failure analysis and a reproducible negative result. A future predictive study should place the target temporally or spatially downstream from the input controls, for example using braking-zone and turn-in features to predict later exit speed or subsequent time loss, instead of using features that define the target.

7. Limitations

- Adaptive segmentation is driver-specific. Segment IDs are not guaranteed to represent the same physical corner across drivers, particularly at Silverstone and Suzuka. This version treats rows as local windows and makes no cross-driver segment-ID claims.
- The lap filter uses a 107% quick-lap threshold with no separate traffic or aborted-lap classifier. Lap deltas extend to 5.566 s, so some comparisons are not clean performance laps.
- Public FastF1 telemetry lacks steering angle, continuous brake pressure, tire temperatures and pressures, suspension channels, fuel load, setup parameters and other team-level data. Brake is Boolean.
- The fastest selected same-driver lap is an extreme-order reference and may embed favorable track evolution or noise. Comparisons across qualifying phases can still contain session-evolution effects.
- Observations are clustered because many comparison laps share the same reference lap. Inferential p-values from row-level correlations are omitted in this revision for that reason.
- The compact delta export is sampled at 50 m even though the analysis grid is 5 m. This naming and storage distinction must be documented in released code.

- The primary model and the earlier sensitivity audit used slightly different feature sets, because `tyre_life_delta` was present in the headline model and absent from the audit baseline. This revision does not mix those metrics.
- The residual experiment is negative and should not be generalized to proprietary high-frequency team telemetry or to other racing series.

8. Conclusion

Across 372 same-driver qualifying-lap comparisons and 2,487 adaptive segment observations from five 2024 Formula 1 circuits, a direct analytic inverse-speed reconstruction achieved mean leave-one-circuit-out MAE of 0.0408 s and mean R^2 of 0.8127, outperforming the best evaluated learned model at 0.0563 s MAE and R^2 0.7258. After subtracting the analytic component, control and powertrain features failed to predict the residual on every held-out circuit, mean R^2 -0.275.

The study does not support the claim that machine learning provides independent cross-circuit predictive value for this target. What it provides instead is an engineering case study in telemetry decomposition, baseline design and methodological self-audit.

The practical lesson is the one worth carrying forward. Before adding machine learning to a motorsport analysis problem, check whether the relevant physics or algebra already solves the task more accurately and more transparently.

9. Data, Code and Reproducibility

The public release should include the extraction script, the analysis script, exact dependency versions, session identifiers, derived feature tables, evaluation tables, the baseline audit and the figures. Raw Formula 1 telemetry should be re-fetched through FastF1 rather than redistributed without confirming upstream rights.

The data source was FastF1 3.8.3. Extraction sessions were 2024 Bahrain Qualifying, Monaco Qualifying, British Grand Prix Qualifying, Italian Grand Prix Qualifying and Japanese Grand Prix Qualifying. Primary analysis seed: 42. Reviewer-audit and residual check seed: 0. Model hyperparameters and selection rules appear in Sections 2 and 4.

10. Research Integrity and AI-Assistance Disclosure

This is independent student research. It is not institutionally supervised or endorsed by New York Institute of Technology.

Generative AI tools, including OpenAI ChatGPT and Anthropic Claude, assisted with code development and debugging, manuscript drafting and methodological review. I executed the data extraction and analysis pipeline locally, supplied the resulting artifacts for audit, reviewed the reported outputs, and accept responsibility for the final content, claims and disclosures. AI systems are not listed as authors.

The manuscript was substantially revised after a critical review identified missing analytic baselines, target coupling, segment-consistency limitations and inconsistent feature-set reporting.

Author contribution: Kazi Sadia Anowar - project conception, local data acquisition, execution of the analysis pipeline, project integration, review of results, manuscript preparation and final accountability.

Conflict of interest: None declared.

11. Revision Note

Version 2.0 replaces the original ML-first interpretation. It adds non-learned baselines, moves the residualized test into the main results, reports fold variation and target-scale normalization, corrects the Silverstone interpretation, defines the lap-selection rule, explains the 5 m analysis grid against the 50 m compact export, discloses the tyre-life feature-set mismatch, reports outlier ranges, and reframes adaptive segments as local windows instead of canonical cross-driver corner IDs.

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Appendix A. Key Reproducibility Numbers

- 469 selected laps; 467 successfully extracted telemetry laps; 372 comparison pairs; 2,487 adaptive segment observations.
- Original configured Gradient Boosting: MAE 0.0563 ± 0.0310 s across circuit folds; mean R^2 0.7258 ± 0.0389 .
- Analytic reconstruction: mean MAE 0.0408 s; mean R^2 0.8127.
- Residual model: mean held-out R^2 -0.275; negative on every circuit.
- Official versus integrated lap delta: $r=0.9937$ overall; mean absolute difference 0.0846 s; bias +0.0119 s; $r=0.9428$ for pairs under 1 s.
- 98.9% of lap pairs use the same tyre compound.